Day-of-the-Week Effects in Cryptocurrency Returns

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the mean daily return of an equal-weight crypto basket is the same on all weekdays; H_1: at least one weekday differs
Data: 3,272 days with at least three assets, 2017-11-10 to 2026-10-25
Method: Weekday dummies with Newey–West HAC errors (7 lags) and a joint Wald test of the six dummies
Results: Wald $\chi^2_6 = 10.70$, $p = 0.0983$; highest mean Sat (0.607%), lowest Thu (0.010%).
Research decision: REJECTED

1 Design

Crypto markets trade every day, so weekday patterns can arise from the weekly rhythm of institutional and retail activity [Caporale and Plastun, 2019]. The test is a single joint hypothesis on all weekdays rather than a search for the best day, which avoids data-snooping across seven comparisons.

2 Results

Table 1: Equal-weight crypto basket return by weekday (UTC)

Weekday	Days	Mean	Std. dev.
Mon	467	0.142%	4.60%
Tue	467	0.106%	4.14%
Wed	467	0.352%	4.85%
Thu	467	0.010%	5.25%
Fri	468	0.506%	4.36%
Sat	468	0.607%	3.37%
Sun	468	0.028%	3.52%

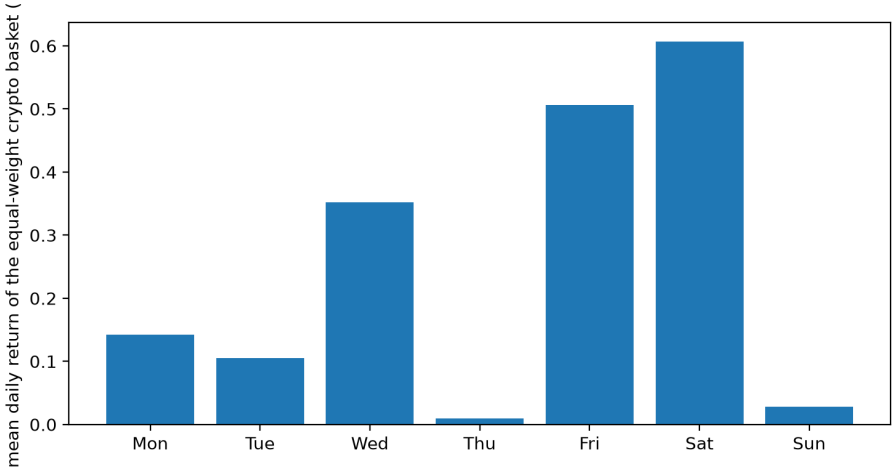


Figure 1: Mean daily return by weekday.

3 Limitations

Daily bars close at 00:00 UTC, so the weekday label follows UTC rather than any local trading day. The basket includes more assets in later years. The decade has a few extreme days, and the joint test is not robust to a single outlier driving one weekday's mean.

References

Guglielmo Maria Caporale and Alex Plastun. Day of the week effect in cryptocurrency markets. *Finance Research Letters*, 31:258–269, 2019.